

IMPLEMENTATION PLAN · TO 7 OCTOBER

Who Owns the Flood · Harvard GSD MDE · 27 Sep 2026

TOOLS

- **PowerShell 5.1** scripts for the whole pipeline (no Python); **Git/GitHub**
- **D3.js v7** + **TopoJSON**: maps, charts, scroll-driven story
- **Headless Edge**: PDF export and screenshots
- **Claude Code**: coding and research agent; sub-agents follow rules (verbatim quotes, URLs, confidence ratings)
- **Data**: Census 2011 language; JRC Global Surface Water; eThekwinini 1:100 floodplain; WMO OSCAR; 84 relocation sites; 199 news articles; 35 world cases

SKILLS AND WORKFLOWS

- Point-in-polygon and buffer queries against the floodplain; satellite pixel sampling
- Random-point baseline (fixed seed); binomial significance test
- Case coding into structured tables; a claim-by-claim evidence ledger
- Second-agent verification; SHA-256 data manifest; one-command rebuilds
- 7-state storyboard, delivered as a scroll story and a PDF brief

STILL TO BUILD

- **Seven whys × 10 cases**: finish the cross-country test (stopped by a usage limit)
- **Reflooding**: search the remaining 54 of 84 sites; confirm the Feb 2025 camp name
- **Why 7**: source apartheid-era siting of Durban townships
- **Join the thesis**: language spoken at each camp, and whether warnings reached it
- **Deck and audio**: warning recordings in isiZulu, English and Tshivenda
- **Accessibility**: screen reader, contrast, isiZulu and Afrikaans captions

TECHNICAL PROBLEMS

- Site coordinates only accurate to **±1–3 km**, so each needs checking on satellite imagery
- **Only Durban** has an open floodplain layer; Cape Town and Gqeberha need floodline data or the JRC hazard map
- The image step is **Windows-only** (System.Drawing); a Mac needs PowerShell 7 or a Python port
- Sources that **block automated fetching** (IOL, ReliefWeb, SAFLII) need manual checks
- PDF and live pages not yet **checked on a phone or by eye**; the links must be shared manually

NEXT 9 DAYS

DAYS	WANDILE	PARTNER
28–29 Sep	Finish the seven-whys test; source why 7	Verify site coordinates on imagery (Durban first)
30 Sep–1 Oct	Reflooding search of 54 sites; IOL check	Cape Town / NMB floodlines; extend the floodplain test
2–3 Oct	Seven-whys page and camp-language layer in the brief	Accessibility and mobile test; record warning audio
4–5 Oct	Presentation deck; rehearse	Rebuild on a clean machine from REPRODUCE.md
6 Oct	Final edits; export and share	Final check of sources and captions
7 Oct	Present	

Rename the owners as you agree. Repository:
github.com/wandilemthiyane94-jpg/mde-dataviz